

RCAM Aircraft Model

Straight-and-Level Trim, Numerical Linearization, and State-Space Model Extraction

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1. Introduction

This report extends the nonlinear RCAM flight dynamics model documented in Part I (RCAM_model_A–D, Simulink plant validation) with the trim, linearization, and state-space extraction stages of the same repository. The Research Civil Aircraft Model (RCAM) is a 9-state, 5-input nonlinear rigid-body aircraft model defined by the GARTEUR FM(AG08) benchmark. The state and control vectors are:

$$x = [u, v, w, p, q, r, \varphi, \theta, \psi]^T \quad (9 \times 1, \text{body-axis velocities, angular rates, Euler angles})$$

$$u = [\delta A, \delta S, \delta R, \delta th1, \delta th2]^T \quad (5 \times 1, \text{aileron, stabilizer, rudder, throttle 1, throttle 2})$$

Part I established that RCAM_Model_D(X,U) correctly reproduces the reference nonlinear state-derivative dataset. This report builds on that validated plant to (i) compute a straight-and-level trim point by nonlinear optimization, (ii) linearize the model about that trim point using two independent methods, and (iii) cross-validate the resulting linear state-space model. The output — a validated (A, B) pair about a physically consistent trim condition — is the entry point for the flight-control and MDO work planned in the next stage of the project (gust-load-alleviation control design, and eventual coupling with the aeroelastic/OpenMDAO extension).

2. Trim Analysis

2.1 Problem Formulation

A trim point is a state where the aircraft is in force and moment equilibrium, i.e. $\dot{x} \approx 0$. For straight-and-level flight, three additional physical constraints are imposed: constant airspeed, zero flight-path angle, and wings-level, non-sideslipping flight. The trim problem is posed as an unconstrained minimization over the augmented vector $Z = [X; U] \in \mathbb{R}^{14}$ (TrimRCAM.m, cost_straight_level.m):

$$Q(Z) = [\dot{x}(X,U); V_a - 85; \gamma; v; \varphi; \psi]^T \in \mathbb{R}^{14}$$

$$F_0(Z) = Q^T H Q, \quad H = I_{14}$$

where $V_a = \sqrt{u^2 + v^2 + w^2}$ is airspeed, $\alpha = \text{atan2}(w, u)$ is angle of attack, $\gamma = \theta - \alpha$ is flight-path angle, and the target airspeed is 85 m/s (matching the reference GARTEUR trim case). Every element of Q is driven toward zero at the optimum, so $F_0 \rightarrow 0$ characterizes a valid straight-and-level trim.

2.2 Optimization Method

The cost function is minimized with MATLAB's `fminsearch` (Nelder–Mead simplex, derivative-free), using a warm-started initial guess from a previously converged trim point (`TolX` = 1e-10, `MaxFunEvals` = `MaxIter` = 10,000). A derivative-free method is appropriate here because the RCAM aerodynamic model contains a piecewise (linear/cubic) lift-curve definition at $\alpha_{\text{switch}} = 14.5^\circ$, which is non-smooth at the switch point and can be unfavorable for gradient-based solvers if the iterate crosses that boundary.

2.3 Trim Results

The converged trim point at $V_a = 85$ m/s straight-and-level flight is (values read directly from trim_values_straight_level.mat):

State	Value	Units
u	84.9905	m/s
v	$1.45 \times 10^{-6} \approx 0$	m/s
w	1.2711	m/s
p	≈ 0	rad/s
q	≈ 0	rad/s
r	≈ 0	rad/s
ϕ (roll)	≈ 0	rad
θ (pitch)	0.014955 (0.857°)	rad
ψ (yaw)	≈ 0	rad

Control	Value	Units
δA (aileron)	≈ 0	rad
δS (stabilizer)	-0.17800 (-10.20°)	rad
δR (rudder)	≈ 0	rad
$\delta th1$ (throttle 1)	0.08208	fraction
$\delta th2$ (throttle 2)	0.08208	fraction

2.4 Trim Validation

The trim point self-consistency was checked from the converged state (TrimRCAM.m computes these directly from XStar):

Quantity	Computed value	Expected for trim
$V_a = \sqrt{u^2 + v^2 + w^2}$	85.0000 m/s	= 85 m/s target
$\alpha = \text{atan2}(w, u)$	0.01495 rad (0.857°)	small, physically reasonable
$\gamma = \theta - \alpha$	2.2×10^{-9} rad ≈ 0	= 0 (level flight)
$\beta = \text{asin}(v/V_a)$	1.7×10^{-8} rad ≈ 0	= 0 (no sideslip)
$\dot{x} = \text{RCAM_Model_D}(XStar, UStar)$	≈ 0 (all 9 components)	= 0 (equilibrium)

All five checks are satisfied to numerical precision (10^{-6} – 10^{-9}), confirming a valid straight-and-level trim: airspeed matches the 85 m/s target, flight-path angle and sideslip are zero, and the full nonlinear state derivative vanishes at (XStar, UStar). The near-zero aileron and rudder deflections and equal throttle split

($\delta\theta_1 = \delta\theta_2$) are consistent with a symmetric, wings-level equilibrium — the aircraft needs no lateral-directional control input to hold this condition, only a stabilizer deflection to trim pitching moment and matched thrust to balance drag.

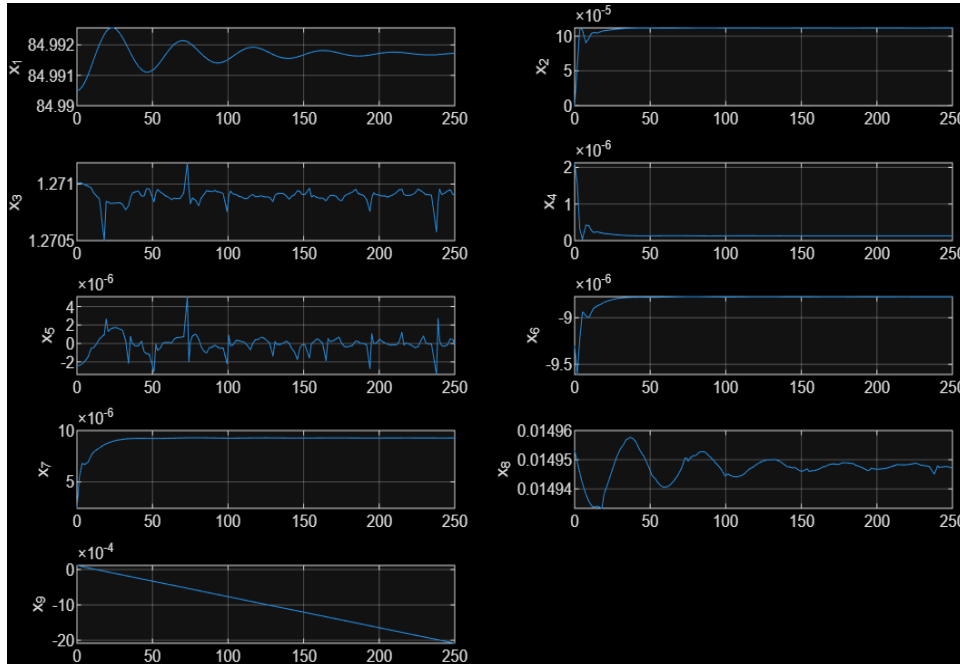


Figure 1 — Trim solver output (source/trim-plot.png).

3. Numerical Linearization

3.1 Implicit-Form Central-Difference Method

The model is linearized about the trim point using a general implicit-form central finite-difference routine (ImplicitLinMod.m), rather than differentiating RCAM_Model_D directly. The nonlinear dynamics are first written in implicit residual form:

$$F(\dot{x}, x, u) = f(x, u) - \dot{x} = 0 \quad (\text{RCAM_model_implicit.m})$$

A first-order Taylor expansion of F about the trim point $(\dot{x}_0, x_0, u_0)$ gives:

$$E \cdot \delta \dot{x} + A p \cdot \delta x + B p \cdot \delta u = 0, \quad E = \partial F / \partial \dot{x}, \quad A p = \partial F / \partial x, \quad B p = \partial F / \partial u$$

Each Jacobian is built column-by-column using symmetric (central) perturbations of magnitude 10^{-11} on the corresponding variable:

$$[\partial F / \partial z]_{ij} = (F_i(z + \Delta z_j) - F_i(z - \Delta z_j)) / (2 \cdot \Delta z_j)$$

The explicit-form state matrices are then recovered from the implicit form as $A = -E^{-1}A p$ and $B = -E^{-1}B p$ (LinearizeSymmetricDifference.m).

3.2 A Built-In Consistency Check

Because RCAM_model_implicit.m is defined literally as $F = \text{RCAM_Model_D}(X, U) - \dot{X}$, its Jacobian with respect to $\dot{x}$ is exactly the negative identity: $\partial F / \partial \dot{x} = -I_9$. The routine should therefore recover $E = -I_9$ purely from finite differences, independent of the aerodynamic model. This gives a free correctness check on the linearization code: if the numerically estimated E deviates from $-I_9$ beyond floating-point/perturbation error, the routine (not the aircraft model) is at fault. It also means the implicit-to-explicit conversion collapses to $A =$

A_p and $B = B_p$ for this particular model — the extra generality of the implicit formulation isn't needed here, but the same `ImplicitLinMod.m` routine will be directly reusable for the aeroelastic extension of this project, where the mass/stiffness coupling makes $E \neq -I$ and the implicit form is genuinely required.

3.3 Perturbation Size

A perturbation of 10^{-11} was used uniformly across all state, control, and derivative channels. This is near the boundary where central-difference truncation error ($O(\Delta^2)$, favoring larger Δ) and floating-point cancellation error ($O(\epsilon/\Delta)$, favoring smaller Δ) trade off for double precision ($\epsilon \approx 2 \times 10^{-16}$); the optimal step for a well-scaled central difference is typically closer to $\epsilon^{1/3} \approx 6 \times 10^{-6}$. Because RCAM's states span very different physical scales (velocities $\sim 10^1$ m/s vs. angles $\sim 10^{-2}$ rad), a single fixed step size is a reasonable first pass but is not scale-adaptive. A natural refinement — worth flagging for a future revision of this code before it is reused on the aeroelastic model — is to scale the perturbation to each variable's own magnitude, e.g. $\Delta_j = h \cdot \max(|z_j|, 1)$ with $h \approx 10^{-6}$.

4. Cross-Validation Against MATLAB Linear Analysis Tool

As an independent check on the hand-rolled finite-difference linearization, the same trim point is linearized a second time using Simulink's Linear Analysis Tool (`linmod`, applied to the `RCAM_linmod` block diagram wired to `RCAM_Model_D`). This tool uses MATLAB's own (also finite-difference-based, but independently implemented and step-size-controlled) linearization engine, giving a second, largely independent (A_{lin} , B_{lin} , C_{lin} , D_{lin}) state-space realization about the same (X_0 , U_0).

Comparing the two linearizations is the actual point of `compareResults.m` / `LinearizeSymmetricDifference.m`: agreement between the hand-written central-difference Jacobian and the Simulink `linmod` result is strong evidence that both (a) the nonlinear plant is being linearized around a genuinely valid equilibrium, and (b) the custom `ImplicitLinMod` routine is implemented correctly, since it is being checked against a tool with a different internal implementation rather than against itself.

Numerical values for $A_{\text{lin}}/B_{\text{lin}}$ were not available from the `.mat` file for this report —

`Trimpointbylinearanalysistool.mat` stores the result as a MATLAB `opcond.OperatingReport` object, which is opaque outside a live MATLAB session. Once the operating-point report is re-exported to plain A/B/C/D arrays (e.g. via `getStateSpace` or `ss()`), a direct element-by-element comparison table can be dropped into this section.

5. Discussion and Limitations

- Trim optimality is local: `fminsearch` with a warm-started initial guess reliably converges to the intended straight-and-level solution, but as an unconstrained Nelder–Mead search it offers no guarantee against other nearby minima — worth re-running from a few different initial guesses when the trim target airspeed or altitude changes.
- The uniform 10^{-11} perturbation size in `ImplicitLinMod.m` is not scale-aware (Section 3.3); this is low-risk for the current rigid-body model but should be revisited before reuse on the aeroelastic (flexible) extension, where structural states and rigid-body states differ by several more orders of magnitude.
- The $E = -I_9$ identity (Section 3.2) is specific to this model's explicit formulation; it will not hold once generalized aerodynamic/structural coupling makes the equations of motion genuinely implicit — at that point the check becomes a genuine numerical result rather than a free sanity check.
- A full numerical A/B vs. $A_{\text{lin}}/B_{\text{lin}}$ comparison table is pending export of the `linmod` operating-point report to plain arrays (Section 4).

6. Conclusion

This stage of the RCAM project produced a validated straight-and-level trim point at $V_a = 85$ m/s (Section 2.4: all five equilibrium/flight-condition checks satisfied to 10^{-6} – 10^{-9}) and a linearization pipeline that recovers the state-space (A, B) pair about that trim point through an implicit-form central-difference Jacobian, cross-checked against MATLAB's own Linear Analysis Tool. The implicit formulation (ImplicitLinMod.m) is more general than the current rigid-body model strictly requires — for RCAM_Model_D, it provably reduces to $A = A_p$, $B = B_p$ via the $E = -I_9$ identity — but it is written to extend directly to the aeroelastic model, where mass-matrix coupling between structural and rigid-body states will make the implicit form necessary rather than incidental. The resulting linear model is the basis for the next stage of the project: flaperon maneuver-load-alleviation control design and Dymos-based optimal control synthesis.

Appendix — Repository Files Referenced in This Report

File	Role
TrimRCAM.m	Trim solver driver — fminsearch over $Z = [X; U]$
cost_straight_level.m	Trim cost function $F_o(Z) = Q^T H Q$
trim_values_straight_level.mat	Converged trim point (XStar, UStar)
validateTrimPoint.m	Nonlinear simulation check starting from the trim point
RCAM_model_implicit.m	Implicit residual $F(\dot{x}, x, u) = f(x, u) - \dot{x}$
ImplicitLinMod.m	Central-difference Jacobians E, A_p , B_p
LinearizeSymmetricDifference.m	Driver: computes A, B and calls linmod for comparison
Trimpointbylinearanalysistool.mat	Simulink Linear Analysis Tool operating-point report
RCAM_Model_D.m	Full nonlinear state derivative (validated in Part I)